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TREND-BASED MOCK PAPER

GATE Computer Science and Information Technology

Mock Paper — 02

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Computer Science and Information Technology
Paper:	Mock 02
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [2-NAT] [2 marks]

Antonym of 'ephemeral':

(Numerical Answer Type — Enter the numerical value)

Q2. [2-NAT] [2 marks]

In a class of 60 students, 30 play cricket, 25 play football, and 15 play both. How many play neither?

(Numerical Answer Type — Enter the numerical value)

Section B: Computer Science and Information Technology

Q3. [3-MCQ] [4 marks]

Using 3 frames, which page replacement algorithm would not cause any page fault for reference string 0,1,2,3,0,1,4,0,1,2,3,4?

- (A) FIFO
- (B) LRU
- (C) Optimal
- (D) None of these

Q4. [2-NAT] [2 marks]

Which of the following is NOT a necessary condition for deadlock?

(Numerical Answer Type — Enter the numerical value)

Q5. [2-MCQ] [2 marks]

Semaphore with value 1 is:

- (A) Counting
- (B) Binary (mutex)
- (C) Monitor
- (D) Lock variable

Q6. [2-MCQ] [2 marks]

TLB stands for:

- (A) Translation Lookaside Buffer
- (B) Time Local Buffer
- (C) Total Logic Block
- (D) None

Q7. [2-MSQ] [2 marks]

Banker's algorithm is for:

- (A) Detection
- (B) Avoidance
- (C) Prevention
- (D) Recovery

Q8. [2-NAT] [2 marks]

Round-robin uses:

(Numerical Answer Type — Enter the numerical value)

Q9. [2-MCQ] [2 marks]

Inode contains:

- (A) File name
- (B) Metadata
- (C) File data
- (D) Directories

Q10. [1-MCQ] [1 mark]

Critical section problem solution requirements:

- (A) Mutual exclusion only
- (B) Progress and bounded waiting also
- (C) No requirements
- (D) Only progress

Q11. [1-MCQ] [1 mark]

Shortest remaining time first (SRTF) is:

- (A) Preemptive SJF
- (B) Non-preemptive SJF
- (C) Round robin
- (D) FCFS

Q12. [3-MCQ] [3 marks]

In a system with 5 frames and reference string 1,2,3,4,1,2,5,1,2,3,4,5. LRU faults:

- (A) 7
- (B) 8
- (C) 9
- (D) 10

Q13. [1-MCQ] [1 mark]

Belady's anomaly (more frames -> more faults) occurs in:

- (A) LRU
- (B) Optimal
- (C) FIFO

(D) LFU

Q14. [2-NAT] [2 marks]

Kruskal uses:

(Numerical Answer Type — Enter the numerical value)

Q15. [2-NAT] [2 marks]

Worst-case time complexity of merge sort:

(Numerical Answer Type — Enter the numerical value)

Q16. [2-MCQ] [2 marks]

LCS uses:

- (A) Greedy
- (B) Divide and Conquer
- (C) Dynamic Programming
- (D) Backtracking

Q17. [2-MSQ] [2 marks]

Time complexity of binary search:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n \log n)$
- (D) $O(1)$

Q18. [3-MCQ] [4 marks]

The time complexity of Dijkstra's algorithm using a binary heap is:

- (A) $O(V^2)$
- (B) $O(E \log V)$
- (C) $O(V \log V + E)$
- (D) $O(V \log V)$

Q19. [2-NAT] [2 marks]

BFS uses:

(Numerical Answer Type — Enter the numerical value)

Q20. [2-MCQ] [2 marks]

Merge sort space:

- (A) $O(1)$
- (B) $O(\log n)$
- (C) $O(n)$
- (D) $O(n^2)$

Q21. [1-MCQ] [1 mark]

Dijkstra from source s . After extracting u from PQ, which is always true?

- (A) $d[u]$ is final
- (B) $d[u]$ is shortest path
- (C) All vertices processed
- (D) $d[u] \leq d[v]$ for all v

Q22. [1-MCQ] [1 mark]

Floyd-Warshall computes all-pairs shortest paths in:

- (A) $O(n^3)$
- (B) $O(n^2 \log n)$
- (C) $O(nm)$
- (D) $O(n^2)$

Q23. [1-MCQ] [1 mark]

Floyd-Warshall computes all-pairs shortest paths in:

- (A) $O(n^3)$
- (B) $O(n^2 \log n)$
- (C) $O(nm)$
- (D) $O(n^2)$

Q24. [1-MCQ] [1 mark]

Minimum spanning tree with Kruskal's on sorted edges:

- (A) Always unique
- (B) May not be unique
- (C) Never unique
- (D) Depends on graph

Q25. [2-MCQ] [2 marks]

Determinant of 2×2 identity:

- (A) 0
- (B) 1
- (C) 2
- (D) -1

Q26. [2-MCQ] [2 marks]

Probability of at least one head in 3 coin tosses:

- (A) $1/8$
- (B) $3/8$
- (C) $7/8$
- (D) $1/2$

Q27. [2-MCQ] [2 marks]

Let A be a 3×3 matrix with eigenvalues 1, 2, and 3. Then the trace of A^2 is:

- (A) 6
- (B) 7
- (C) 14
- (D) 15

Q28. [2-MCQ] [2 marks]

Mean of binomial $n=10, p=0.5$:

- (A) 2
- (B) 5
- (C) 10
- (D) 0.5

Q29. [2-NAT] [2 marks]

Derivative of x^3 at $x=2$:

(Numerical Answer Type — Enter the numerical value)

Q30. [1-NAT] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

(Numerical Answer Type — Enter the numerical value)

Q31. [1-MCQ] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
 - (B) $1/(1-z)$
 - (C) $\ln(1-z)$
 - (D) $e^{-\lambda}(1+z)$
-

Q32. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
 - (B) 8
 - (C) 16
 - (D) 12
-

Q33. [1-MCQ] [1 mark]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
 - (B) $\|u\| + \|v\|$
 - (C) $\|u\| / \|v\|$
 - (D) $\sqrt{\|u\| * \|v\|}$
-

Q34. [1-NAT] [1 mark]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

(Numerical Answer Type — Enter the numerical value)

Q35. [1-MCQ] [1 mark]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
 - (B) $1/(s-2)^2$
 - (C) $s/(s+2)^2$
 - (D) $2/(s+2)^2$
-

Q36. [2-MCQ] [2 marks]

Which protocol operates at the application layer of the TCP/IP model?

- (A) TCP
 - (B) UDP
 - (C) HTTP
 - (D) IP
-

Q37. [2-MCQ] [2 marks]

CSMA/CD is used in:

- (A) Token Ring

- (B) Ethernet
- (C) WiFi
- (D) Bluetooth

Q38. [2-MCQ] [2 marks]

OSPF is:

- (A) Distance vector
- (B) Link state
- (C) Path vector
- (D) Hybrid

Q39. [2-MCQ] [2 marks]

Protocol to find MAC from IP:

- (A) DNS
- (B) ARP
- (C) DHCP
- (D) ICMP

Q40. [1-MCQ] [1 mark]

TCP connection establishment uses:

- (A) 2-way
- (B) 3-way handshake
- (C) 4-way
- (D) 1-way

Q41. [1-MCQ] [1 mark]

Asymmetric encryption:

- (A) AES
- (B) DES
- (C) RSA
- (D) 3DES

Q42. [1-NAT] [1 mark]

CIDR block 200.10.0.0/20 contains how many class C addresses?

(Numerical Answer Type — Enter the numerical value)

Q43. [1-MCQ] [1 mark]

Which uses Bellman-Ford?

- (A) OSPF
- (B) RIP
- (C) BGP
- (D) IS-IS

Q44. [1-NAT] [1 mark]

TCP seq=1000, 500 bytes data. Next expected seq:

(Numerical Answer Type — Enter the numerical value)

Q45. [1-NAT] [1 mark]

In Go-Back-N, sender window size = 7. Sequence numbers needed:

(Numerical Answer Type — Enter the numerical value)

Q46. [2-NAT] [2 marks]

Infix to postfix uses:

(Numerical Answer Type — Enter the numerical value)

Q47. [2-MCQ] [2 marks]

Avg hash lookup:

- (A) $O(n)$
 - (B) $O(\log n)$
 - (C) $O(1)$
 - (D) $O(n^2)$
-

Q48. [2-MCQ] [2 marks]

Inorder traversal of BST gives:

- (A) Sorted order
 - (B) Reverse sorted
 - (C) Level order
 - (D) Random
-

Q49. [2-MCQ] [2 marks]

What is the time complexity of inserting at the beginning of a linked list?

- (A) $O(n)$
 - (B) $O(1)$
 - (C) $O(\log n)$
 - (D) $O(n^2)$
-

Q50. [1-MCQ] [1 mark]

In ER model, relationship set for many-to-many is represented as:

- (A) Table with foreign keys
 - (B) Separate relation table
 - (C) Single relation
 - (D) No table needed
-

Q51. [1-MCQ] [1 mark]

B+ tree leaf nodes are linked for:

- (A) Faster insert
 - (B) Range queries
 - (C) Faster delete
 - (D) Balance
-

Q52. [1-NAT] [1 mark]

R(A,B,C,D) with FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$. Candidate keys:

(Numerical Answer Type — Enter the numerical value)

Q53. [1-MCQ] [1 mark]

Which concurrency protocol avoids cascading aborts?

- (A) 2PL
- (B) Strict 2PL
- (C) Timestamp ordering
- (D) Optimistic

Q54. [1-NAT] [1 mark]

Flip-flop stores:

(Numerical Answer Type — Enter the numerical value)

Q55. [1-NAT] [1 mark]

Full adder has how many inputs:

(Numerical Answer Type — Enter the numerical value)

Q56. [1-MCQ] [1 mark]

The number of Boolean functions of n variables is:

- (A) 2^n
 - (B) $2^{(2^n)}$
 - (C) $2^{(n^2)}$
 - (D) n^2
-

Q57. [1-MCQ] [1 mark]

Which language is regular?

- (A) Equal 0s and 1s
 - (B) Strings ending in ab
 - (C) Palindromes
 - (D) $a^n b^n$
-

Q58. [1-NAT] [1 mark]

PDA recognizes:

(Numerical Answer Type — Enter the numerical value)

Q59. [1-MCQ] [1 mark]

Halting problem:

- (A) Decidable
 - (B) Undecidable
 - (C) Semi-decidable
 - (D) None
-

Q60. [1-MCQ] [1 mark]

Interrupt-driven I/O vs DMA:

- (A) DMA is faster for single byte
 - (B) DMA is better for block transfer
 - (C) Interrupt is better for large data
 - (D) Both same
-

Q61. [1-MCQ] [1 mark]

2K x 8 ROM has how many address lines?

- (A) 2
 - (B) 8
 - (C) 11
 - (D) 2048
-

Q62. [1-MCQ] [1 mark]

Little-endian stores 0x12345678 at addr 100 as:

- (A) 78 at 100, 56 at 101, ...

- (B) 12 at 100, 34 at 101, ...
- (C) 78 at 103, 56 at 102, ...
- (D) 56 at 100, 78 at 101, ...

Q63. [1-MCQ] [1 mark]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q64. [1-MCQ] [1 mark]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q65. [1-NAT] [1 mark]

Lexical analyzer outputs:

(Numerical Answer Type — Enter the numerical value)

ANSWER KEY

Q. No.	Type	Answer	Marks
1	2NAT	B	2
2	2NAT	A	2
3	2MCQ	B	2
4	2MCQ	C	2
5	2MCQ	C	2
6	2MCQ	B	2
7	2NAT	B	2
8	2NAT	B	2
9	2MCQ	C	2
10	2MCQ	A	2
11	2MCQ	B	2
12	2NAT	C	2
13	2NAT	B	2

14	2MCQ	C	2
15	2MSQ	B	2
16	3MCQ	C	4
17	2NAT	B	2
18	2MCQ	C	2
19	3MCQ	C	4
20	2NAT	C	2
21	2MCQ	B	2
22	2MCQ	A	2
23	2MSQ	B	2
24	2NAT	A	2
25	2MCQ	B	2
26	2MCQ	C	2
27	2MCQ	B	2
28	2MCQ	B	2

29	2MCQ	B	2
30	1MCQ	B	1
31	1MCQ	C	1
32	1NAT	A	1
33	1NAT	B	1
34	1MCQ	B	1
35	1MCQ	B	1
36	1NAT	B	1
37	1MCQ	B	1
38	1NAT	A	1
39	1NAT	A	1
40	1MCQ	B	1
41	1MCQ	B	1
42	1MCQ	A	1
43	1MCQ	A	1

44	1MCQ	B	1
45	1MCQ	B	1
46	1MCQ	A	1
47	1MCQ	C	1
48	1NAT	B	1
49	3MCQ	D	3
50	1MCQ	C	1
51	1NAT	B	1
52	1MCQ	A	1
53	1MCQ	C	1
54	1MCQ	B	1
55	1MCQ	A	1
56	1MCQ	A	1
57	1NAT	B	1
58	1MCQ	B	1

59	1NAT	B	1
60	1MCQ	A	1
61	1MCQ	A	1
62	1NAT	C	1
63	1MCQ	A	1
64	1MCQ	B	1
65	1MCQ	A	1

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: easy

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: difficult

Q4.

Answer: C

Solution:

Correct Answer: C

Subject: Engineering Mathematics | Topic: Probability | Difficulty: easy

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: moderate

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Probability | Difficulty: easy

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Programming and Data Structures | Topic: Stacks | Difficulty: easy

Q9.

Answer: C

Solution:

Correct Answer: C

Subject: Programming and Data Structures | Topic: Hashing | Difficulty: easy

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Programming and Data Structures | Topic: Trees | Difficulty: moderate

Q11.

Answer: B

Solution:

Correct Answer: B

Subject: Programming and Data Structures | Topic: Linked List | Difficulty: easy

Q12.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Greedy | Difficulty: moderate

Q13.

Answer: B

Solution:

Correct Answer: B

Subject: Algorithms | Topic: Sorting | Difficulty: moderate

Q14.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Dynamic Programming | Difficulty: easy

Q15.

Answer: B

Solution:

Correct Answer: B

Subject: Algorithms | Topic: Complexity Analysis | Difficulty: easy

Q16.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Graph Algorithms | Difficulty: difficult

Q17.

Answer: B

Solution:

Correct Answer: B

Subject: Algorithms | Topic: Graph Algorithms | Difficulty: moderate

Q18.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Divide and Conquer | Difficulty: moderate

Q19.

Answer: C

Solution:

Correct Answer: C

Subject: Operating Systems | Topic: Memory Management | Difficulty: moderate

Q20.

Answer: C

Solution:

Correct Answer: C

Subject: Operating Systems | Topic: Deadlocks | Difficulty: easy

Q21.

Answer: B

Solution:

Correct Answer: B

Subject: Operating Systems | Topic: Synchronization | Difficulty: moderate

Q22.

Answer: A

Solution:

Correct Answer: A

Subject: Operating Systems | Topic: Virtual Memory | Difficulty: moderate

Q23.

Answer: B

Solution:

Correct Answer: B

Subject: Operating Systems | Topic: Deadlocks | Difficulty: easy

Q24.

Answer: A

Solution:

Correct Answer: A

Subject: Operating Systems | Topic: Process Scheduling | Difficulty: moderate

Q25.

Answer: B

Solution:

Correct Answer: B

Subject: Operating Systems | Topic: File Systems | Difficulty: difficult

Q26.

Answer: C

Solution:

Correct Answer: C

Subject: Computer Networks | Topic: Application Layer | Difficulty: moderate

Q27.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Data Link Layer | Difficulty: easy

Q28.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Routing | Difficulty: easy

Q29.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Network Layer | Difficulty: difficult

Q30.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Transport Layer | Difficulty: easy

Q31.

Answer: C

Solution:

Correct Answer: C

Subject: Computer Networks | Topic: Security | Difficulty: moderate

Q32.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Logic | Topic: Sequential Circuits | Difficulty: moderate

Q33.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Logic | Topic: Combinational Circuits | Difficulty: easy

Q34.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Logic | Topic: Boolean Algebra | Difficulty: easy

Q35.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Computation | Topic: Regular Languages | Difficulty: easy

Q36.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Computation | Topic: Context Free Languages | Difficulty: moderate

Q37.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Computation | Topic: Decidability | Difficulty: moderate

Q38.

Answer: A

Solution:

Correct Answer: A

Subject: Compiler Design | Topic: Lexical Analysis | Difficulty: moderate

Q39.

Answer: A

Solution:

/20 spans $2^{(24-20)} = 16$ class C (/24) networks.

Subject: Computer Networks | Topic: Computer Networks | Difficulty: moderate

Q40.

Answer: B

Solution:

M:N relationship requires separate relation table with foreign keys to both entity tables.

Subject: DBMS | Topic: DBMS | Difficulty: easy

Q41.

Answer: B

Solution:

B+ tree: all data at leaves, linked sequentially. Enables efficient range scans without root-to-leaf traversal.

Subject: DBMS | Topic: DBMS | Difficulty: moderate

Q42.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3+1/x^3 = 27-3*3 = 18$.

Subject: Data Structures | Topic: Data Structures | Difficulty: moderate

Q43.

Answer: A

Solution:

Dijkstra extracts in non-decreasing distance order. Once extracted, $d[u]$ is finalized (non-negative edges).

Subject: Algorithms | Topic: Algorithms | Difficulty: easy

Q44.

Answer: B

Solution:

CS requirements: (1) Mutual exclusion, (2) Progress (no deadlock), (3) Bounded waiting (no starvation).

Subject: Operating Systems | Topic: Operating Systems | Difficulty: difficult

Q45.

Answer: B

Solution:

DMA: direct memory access. Better for block transfers (no CPU per byte). Interrupt per byte.

Subject: Computer Organization | Topic: Computer Organization | Difficulty: easy

Q46.

Answer: A

Solution:

SRTF: preemptive version of SJF. Preempts if new process has shorter remaining time.

Subject: Operating Systems | Topic: Operating Systems | Difficulty: moderate

Q47.

Answer: C

Solution:

2K = 2048 locations. Need $\log_2 = 11$ address lines.

Subject: Computer Organization | Topic: Computer Organization | Difficulty: easy

Q48.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q49.

Answer: D

Solution:

LRU with 5 frames: all first 5 distinct are faults (5). Then 1,2 present. 5 new (replace 3,4,1,2,3). 4 new (replace 1,2,3,4). 5 new. Total: $5+1+1+1+1+1=10$.

Subject: Operating Systems | Topic: Operating Systems | Difficulty: moderate

Q50.

Answer: C

Solution:

Belady's anomaly: FIFO can have more faults with more frames. LRU and Optimal are stack algorithms (no anomaly).

Subject: Operating Systems | Topic: Operating Systems | Difficulty: moderate

Q51.

Answer: B

Solution:

Closures: $(AB)^+=ABCD$, $(BD)^+=ABCD$, $(BC)^+=ABCD$. Minimal: AB and BD.

Subject: DBMS | Topic: DBMS | Difficulty: moderate

Q52.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q53.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q54.

Answer: B

Solution:

RIP: distance vector, Bellman-Ford. OSPF: link-state (Dijkstra).

Subject: Computer Networks | Topic: Computer Networks | Difficulty: difficult

Q55.

Answer: A

Solution:

Floyd-Warshall: $O(n^3)$ DP over all intermediate vertices.

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q56.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q57.

Answer: B

Solution:

Next expected = $1000 + 500 = 1500$.

Subject: Computer Networks | Topic: Computer Networks | Difficulty: moderate

Q58.

Answer: B

Solution:

Strict 2PL: holds all locks until commit. Prevents dirty reads, cascading aborts, and unrepeatable reads.

Subject: DBMS | Topic: DBMS | Difficulty: easy

Q59.

Answer: B

Solution:

Window = N : need $N+1$ sequence numbers to distinguish. For $N=7$: need 8 (3 bits).

Subject: Computer Networks | Topic: Computer Networks | Difficulty: difficult

Q60.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3 + 1/x^3 = 27 - 3*3 = 18$.

Subject: Data Structures | Topic: Data Structures | Difficulty: moderate

Q61.

Answer: A

Solution:

Little-endian: LSB first. 0x78 at 100, 0x56 at 101, 0x34 at 102, 0x12 at 103.

Subject: Computer Organization | Topic: Computer Organization | Difficulty: difficult

Q62.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q63.

Answer: A

Solution:

Floyd-Warshall: $O(n^3)$ DP over all intermediate vertices.

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q64.

Answer: B

Solution:

If edge weights are equal, multiple MSTs possible. Kruskal picks arbitrarily among equal-weight edges.

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q65.

Answer: A

Solution:

$L\{t*f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Operating Systems: 11 questions, 22 marks
Algorithms: 11 questions, 20 marks
Engineering Mathematics: 11 questions, 16 marks
Computer Networks: 10 questions, 14 marks
Programming and Data Structures: 4 questions, 8 marks
General Aptitude: 2 questions, 4 marks
DBMS: 4 questions, 4 marks
Digital Logic: 3 questions, 3 marks
Theory of Computation: 3 questions, 3 marks
Computer Organization: 3 questions, 3 marks
Data Structures: 2 questions, 2 marks
Compiler Design: 1 questions, 1 marks

Difficulty Distribution:

Easy: 24 | Moderate: 33 | Difficult: 8

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.